

Code generation

Write a Python function to calculate the factorial of a given number. Include an explanation and example output.

```
def factorial(n):  
    if n < 0:  
        return "Factorial is not defined for negative numbers"  
  
    result = 1  
    for i in range(1, n + 1):  
        result *= i  
  
    return result
```

Example

```
num = 5  
print("Factorial of", num, "is", factorial(num))
```

Code translation

Translate the following Python code into JavaScript:

```
def greet(name):  
    print("Hello," + name)
```

Here is the equivalent JavaScript code:

```
// JavaScript  
  
function greet(name) {  
    console.log("Hello," + name);  
}  
  
// Example  
greet("Shalini");
```

Output:

```
Hello, Shalini
```

The Python `def` becomes `function` in JavaScript, and `print()` becomes `console.log()`.

Code optimization

Optimize the following Python code to reduce execution time and explain the improvements:

Example: Optimizing Python Code

Before optimization:

```
Python   Run
```

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

even_numbers = []
for num in numbers:
    if num % 2 == 0:
        even_numbers.append(num)

print(even_numbers)
```

Optimized code:

```
Python   Run
```

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

even_numbers = [num for num in numbers if num % 2 == 0]

print(even_numbers)
```

Output:

```
[2, 4, 6, 8, 10] 
```

Short explanation:

The original code uses a separate loop and `append()`. The optimized version uses **list comprehension**, which is shorter, cleaner, and generally faster for creating a list from another list.

Code debugging

Debug the following Java code and explain why it throws a `NullPointerException`:

```
String name = null;  
System.out.println(name.length());
```

Code Debugging – Java

Buggy code:

```
Java  
  
String name = null;  
  
System.out.println(name.length());
```

Why does it throw `NullPointerException`?

`name` is assigned `null`, which means it does not refer to any `String` object. Calling `name.length()` tries to access a method on `null`, causing a `NullPointerException`.

Corrected code:

```
Java  
  
String name = "Shalini";  
  
System.out.println(name.length());
```

Output:

```
7
```

Short explanation:

Always make sure the object is not `null` before calling its methods.